CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) — CHAPTER 4
ANALYSE DATA USING SCENARIO AND GOAL SEEK

Comprehensive Detailed Theory • Unabridged Practical Procedures • Solved Board PYQs & Case Studies

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1. Introduction to Data Analysis Tools in LibreOffice Calc

Spreadsheet applications are not merely used for recording tabular data; they serve as critical decision-support systems for businesses, accountants, and planners. Decision-makers frequently need to evaluate the impact of changing operational conditions (e.g., changes in costs, interest rates, or sales targets) or calculate backward to determine the exact resources needed to achieve a target financial benchmark. LibreOffice Calc provides four robust analysis features:

- **Data Consolidation:** Combines scattered datasets from multiple worksheets or workbooks into a single master summary sheet while applying mathematical/statistical functions.
- **Subtotals:** Groups related data rows sharing identical categorical attributes and computes automated subtotals (sum, average, count) along with an overall grand total.
- **What-If Analysis (Scenarios & Multiple Operations):** Tests alternative hypotheses by altering specific input variables in formulas to simulate and forecast diverse business outcomes.
- **Goal Seek & Solver:** Reverse calculation (back-solving) tools where the target output is fixed, and Calc automatically solves for the required input variable(s) or constraint conditions.

2. Data Consolidation in LibreOffice Calc

Definition: Data Consolidation is the process of collecting, aggregating, and summarizing data from multiple source ranges (located across different worksheets or even separate workbook files) into a single designated destination range, while simultaneously applying a specified summary function (such as Sum, Average, Count, Max, Min).

Core Purposes & Benefits of Consolidation:

- **Facilitates Aggregate Viewing & Analysis:** Enables managers to identify broad trends, multi-branch revenue summaries, and high-level relationships without navigating individual granular sheets.
- **Cross-Workbook & Cross-Sheet Aggregation:** Merges records from multiple departmental files (e.g., *Branch1*, *Branch2*, *Branch3*) into a unified corporate overview.
- **Flexible Matching Mechanisms:** Does not strictly require identical table structures across all sheets; consolidation can match entries dynamically using row headers and column headers.
- **Dynamic Automation:** When enabled with source linking, any retroactive adjustments in branch records automatically reflect in the consolidated output without manual recalculation.

Prerequisites Before Consolidating Data:

- **Uniform Labeling / Header Standardization:** Column headers and row labels must have identical naming across all source worksheets so Calc can match corresponding data points accurately.
- **Consistent Data Types:** Values under matching headers must be numeric and structured consistently across all source sheets.
- **Open Workbooks / Established Access:** All source workbooks must be saved, accessible, and preferably open in Calc if creating dynamic references.

Consolidate Dialog Box — Structural Components & Options:

Dialog Element	Location / Type	Technical Purpose & CBSE Exam Nuance
Function Box	Top drop-down	Selects the aggregation formula. **Sum is the default function**. Available: Average, Count, Max, Min, Product, StDev, Var.
Consolidation ranges	Upper central listbox	Displays all source ranges currently added for consolidation across various worksheets.
Source data ranges	Lower input fields	Allows selecting a predefined *Named Range* from the drop-down or clicking the Shrink button to highlight a range with the mouse.
Copy results to	Target range field	Specifies the upper-left starting cell of the destination sheet where the consolidated table will appear.
Row / Column labels	Under Options (+)	Checkboxes to consolidate data using row headers, column headers, or both. Essential when layouts differ across sheets.
Link to source data	Under Options (+)	**Crucial CBSE Concept:** When checked, creates dynamic formula links so modifications in source sheets automatically update the consolidated sheet.

AS NOTES IT-402

Unabridged Step-by-Step Procedure to Consolidate Data:

Step 1: Open the spreadsheet containing the source data sheets (e.g., *Sheet1*, *Sheet2*, *Sheet3*) and create/select a target worksheet (e.g., *Sheet4*).

Step 2: Navigate to the Menu bar and choose **Data** → **Consolidate...**. The *Consolidate* dialog box opens on the screen.

Step 3: In the **Function** drop-down box at the top, select the required statistical operation (by default, **Sum** is selected).

Step 4: Specify the first source range under **Source data ranges**: click inside the text box (or click the Shrink button), navigate to *Sheet1*, drag across range *B4:B10* (or select an existing named range from the drop-down), and click the **Add** button. The range will appear in the *Consolidation ranges* listbox.

Step 5: Repeat the selection for additional ranges: switch to *Sheet2*, select range *B4:B10*, and click **Add**. Then switch to *Sheet3*, select range *B4:B10*, and click **Add**.

Step 6: Set the target destination: click in the **Copy results to** field, click on the target worksheet (*Sheet4*), and select the target starting cell (e.g., *B4*).

Step 7: Expand the **Options** section by clicking the plus sign (**+**): check **Row labels** and/or **Column labels** if your tables contain text headings. Check **Link to source data** if you require the summary sheet to automatically update whenever source values change.

Step 8: Click **OK**. Calc performs the multi-sheet aggregation and populates the result starting from the designated target cell.

3. Groups and Subtotals in LibreOffice Calc

Definition: The Subtotals tool in Calc automatically groups rows of related data sharing identical values in a designated categorical column, computes intermediate mathematical subtotals (such as Sum, Average, Count) for numeric fields, and creates an overall Grand Total at the bottom.

- **Mandatory Sorting Requirement:** The data range *must be sorted* by the grouping column before applying Subtotals. If the data is unsorted, Calc will create fragmented subtotals every time a duplicate value reappears in non-contiguous rows.
- **Column Labels Required:** The dataset must have distinct column headers in row 1 so Calc can populate the *Group by* and *Calculate subtotals for* fields.
- **Maximum 3 Nesting Levels:** Calc supports up to **3 levels of grouping** (represented by the *1st Group*, *2nd Group*, and *3rd Group* tabs in the Subtotals dialog box).
- **Automatic Outline Margins:** Calc generates an interactive outline panel on the left margin with level buttons (**1, 2, 3**) and plus/minus (**+/-**) toggles to collapse or expand detailed data rows.
- **Automatic Recalculation:** When values in the calculated numeric columns are edited, Calc automatically recalculates the subtotals and grand totals.

Unabridged Step-by-Step Procedure to Calculate Subtotals:

Step 1: Ensure the source data contains proper column headings and is pre-sorted according to the primary category to be grouped (e.g., sorted alphabetically by *Manufacturer* or *Region*).

Step 2: Highlight the entire dataset range, including all column headers and data rows.

Step 3: Click **Data** → **Subtotals...** from the Menu bar. The *Subtotals* dialog box appears.

Step 4: On the **1st Group** tab, select the primary grouping field from the **Group by** drop-down list (e.g., select *Manufacturer*).

Step 5: In the **Calculate subtotals for** listbox, check the checkboxes for the numeric column(s) you wish to calculate (e.g., check *Sales* or *Expenses*).

Step 6: In the **Use function** listbox on the right, select the mathematical function to apply (e.g., **Sum**, **Average**, or **Count**).

Step 7: (Optional) If multi-level nested subtotals are required, switch to the **2nd Group** or **3rd Group** tabs, select a secondary field (e.g., *Model*), select columns to calculate, and choose a function.

Step 8: Click **OK**. Calc inserts grouped subtotal rows with bold labels (e.g., *BMW Sum*, *Audi Sum*) and appends a final **Grand Total** row, activating outline folding controls along the left edge.

4. What-If Analysis: Scenarios

Definition: What-If Analysis is the process of changing the values in worksheet input cells to observe and forecast how those changes will affect the outcome of dependent formula cells. A **Scenario** is a saved package of alternative input values (such as interest rates, sales forecasts, or loan durations) that can be named, stored, and instantly substituted into the worksheet to evaluate different business outcomes.

- **Named Models for Easy Comparison:** Each scenario is assigned a descriptive name (e.g., *Car_Loan_4yr*, *Car_Loan_5yr*, *Best_Case*, *Worst_Case*).
- **Instant Switching:** Scenarios can be toggled using the drop-down title bar directly above the cell range on the sheet or through the **Navigator (F5)** pane.

A.S. NOTES IT-402

- **Border Highlighting:** Each scenario can be customized with a distinct coloured border for visual tracking.
- **Minimum Selection:** To create a scenario, you must select at least **two cells** that will change between scenarios.
- **Copy Back Setting Nuance:** The *Copy back* checkbox (under Settings) copies any manual edits made to values in the scenario range back into the active scenario definition. **CBSE Exam Note:** For static budget models where scenarios should remain locked and uncorrupted, uncheck or clear the *Copy back* box.

Unabridged Step-by-Step Procedure to Create & Manage Scenarios:

Step 1: Set up your base financial model in Calc with input cells (e.g., *Loan Amount*, *Number of Instalments*, *Rate of Interest*) and dependent formula cells (e.g., EMI calculated via =ABS(PMT(Rate/12, Instalments, Principal))).

Step 2: Highlight the input cells whose values will vary between scenarios (hold **Ctrl** while clicking if cells are non-contiguous). You must select at least two cells (e.g., range *B4:B10*).

Step 3: Choose **Tools** → **Scenarios...** from the Menu bar. The *Create Scenario* dialog box opens.

Step 4: In the **Name of Scenario** field, type a clear, descriptive name (e.g., *Car_Loan_4yr*).

Step 5: In the **Comment** box, add optional descriptive notes or author details.

Step 6: Under **Settings**: ensure **Display border** is checked and select a distinct border colour. **Uncheck 'Copy back'** to prevent inadvertent overwrite of scenario values. Check **Prevent changes** if you wish to write-protect the scenario.

Step 7: Click **OK**. The scenario is saved and active, enclosed within a coloured border having a title bar with a drop-down arrow.

Step 8: To create the next alternative scenario, select the same range (*B4:B10*), modify the input values (e.g., change instalments from 48 to 60), select **Tools** → **Scenarios...**, enter a second name (e.g., *Car_Loan_5yr*), uncheck *Copy back*, and click **OK**.

Step 9: To toggle between scenarios, click the drop-down arrow on the scenario border header on the sheet or double-click the scenario name under the Scenarios icon in the **Navigator (F5)**.

5. What-If Analysis: Multiple Operations

Definition: The Multiple Operations tool in LibreOffice Calc is a what-if analysis feature that generates a formula array table to calculate and display results for multiple alternative input values simultaneously. Unlike Scenarios (which display one scenario at a time on the sheet), Multiple Operations lays out a side-by-side comparative grid.

Single-Variable vs. Dual-Variable Tables:

- **One-Variable Table:** Computes one or more formulas for a series of alternative values entered in either a single column OR a single row.
- **Two-Variable Table:** Computes a single formula across a matrix formed by **two intersecting cell arrays**: one set of alternative values placed in a column (e.g., interest rates: 9%, 10%, 11%, 13%, 14%) and a second set placed in a row just above (e.g., durations: 72, 84, 96 months).

Unabridged Step-by-Step Procedure for Multiple Operations (Dual Array):

Step 1: Set up the original formula in the spreadsheet (e.g., EMI formula in cell *E5* referencing loan principal in *B4*, instalments in *B5*, and interest rate in *B6*).

Step 2: Construct the alternative input value matrix: enter one set of input variables in a vertical column (e.g., interest rates in column *F3:F8*) and enter the second set of variables horizontally in a row immediately above (e.g., instalments in row *G2:I2*). Ensure the row and column values do not overlap.

Step 3: Select the entire matrix range containing both row and column input values and the blank intersection cells (e.g., select *F2:I8*).

Step 4: Choose **Data** → **Multiple Operations...** from the Menu bar. The *Multiple Operations* dialog box opens.

Step 5: In the **Formulas** field, enter or click the cell containing the original formula (e.g., E5).

Step 6: In the **Row input cell** field, enter the original cell reference that corresponds to the values listed horizontally in the row (e.g., B5 for instalments).

Step 7: In the **Column input cell** field, enter the original cell reference that corresponds to the values listed vertically in the column (e.g., B6 for interest rate).

Step 8: Click **OK**. Calc populates the entire matrix array with calculated formula results.

6. Goal Seek (Back-Solving Tool)

Definition: Goal Seek is a what-if analysis tool that performs **backward calculation (back-solving)**. When you know the desired target outcome of a formula but do not know the exact input value required to produce that result, Goal Seek calculates backward to find that exact input.

- **Single Variable Only:** Goal Seek can alter **only one input cell** at a time to satisfy a single formula output.

AS NOTES IT-402

- **Direct Formula Dependency:** The formula cell *must directly or indirectly reference* the variable cell being adjusted; otherwise, Calc cannot solve the relationship.
- **Menu Location:** Located under **Tools** → **Goal Seek...**.
- **Solution Prompt:** When a mathematical solution is discovered, Calc displays a dialog stating *'Goal Seek succeeded. Result: [Value]. Insert the result into the variable cell?'*. Clicking **Yes** replaces the original value in the sheet; clicking **No** leaves the sheet unaltered.
- **Error Handling:** If Goal Seek cannot find a feasible mathematical solution (e.g., division by zero or non-converging values), it returns an error notification.

The Three Input Fields in Goal Seek Dialog:

1. Formula cell: The cell reference containing the formula whose outcome you wish to set (e.g., E5 representing EMI, or a cell calculating student total percentage).

2. Target value: The exact numeric value you want the formula to achieve (e.g., 9800 for EMI, or 75 for a target aggregate percentage). Entered as a constant number, not a formula.

3. Variable cell: The cell reference containing the input parameter that Calc is permitted to adjust to reach the target (e.g., B5 for loan duration, or the cell containing Geography marks).

Unabridged Step-by-Step Procedure to Use Goal Seek:

Step 1: Set up the worksheet with the input variable cell and the dependent formula cell (e.g., Formula in *E5* calculating monthly car EMI from principal *B4*, term *B5*, and rate *B6*).

Step 2: Choose **Tools** → **Goal Seek...** from the Menu bar. The *Goal Seek* dialog box appears.

Step 3: Click inside the **Formula cell** field and click on the cell in the worksheet containing the formula (e.g., *E5*).

Step 4: Click inside the **Target value** field and manually enter the target numeric outcome you wish to achieve (e.g., type 9800 or 12000).

Step 5: Click inside the **Variable cell** field and click on the sheet cell whose value Calc must modify to meet the target (e.g., *B5* for number of instalments).

Step 6: Click **OK**. Calc runs iterative calculations.

Step 7: A confirmation dialog appears stating *'Goal Seek succeeded. Result: [Value]. Insert the result into the variable cell?'*. Click **Yes** to overwrite the cell with the solution, or **No** to cancel.

7. Solver Tool (Advanced Optimization Analysis)

Definition: Solver is an advanced, elaborate extension of the Goal Seek tool designed to handle complex optimization equations with **multiple unknown variables** and **multiple operational constraints (limiting conditions)**. While Goal Seek is restricted to finding an exact target for a single input variable, Solver searches for the optimal combination of multiple input variables to **maximize** or **minimize** a target formula result.

- **Objective:** Maximizing profit, minimizing production expenses, optimizing transport routes, or budgeting resources.
- **Variables:** Adjusts multiple cells simultaneously across the spreadsheet.
- **Limiting Constraints:** Accommodates realistic boundaries (e.g., available raw materials, storage capacity, labour hours, non-negative quantities).
- **Menu Location:** Found under **Tools** → **Solver...**.

8. Master Comparison: Goal Seek vs. Solver (High-Yield CBSE 2/4-Marker)

Feature / Aspect	Goal Seek	Solver
Primary Purpose	Calculates backward to find a specific input value for a known single target output.	Optimizes an outcome by finding the **maximum or minimum** value subject to constraints.
Number of Variables	Limited strictly to **one variable cell**.	Can alter **multiple unknown variable cells** simultaneously.
Constraints Handling	Cannot handle operational constraints or limitations.	Applies **multiple constraints** (e.g., material limits, capacity, integer rules).
Problem Complexity	Simple algebraic back-solving (e.g., required marks in 5th subject).	Complex linear / non-linear optimization (e.g., factory profit maximization).
Menu Path	Tools → Goal Seek...	Tools → Solver...

A.S. NOTES IT-402

9. CBSE Board Examination Question Bank & Model Marking Scheme Solutions

Part A: Objective & Multiple Choice Questions (1 Mark Each)

Q1. Which function takes data from a series of worksheets or workbooks and summarises it into a single worksheet that you can update easily? [CBSE SQP 2022-23]
(a) Data Combination (b) Data Merging (c) Data Consolidation (d) Data Concatenation

Ans: (c) Data Consolidation

Q2. Which one of the following is the default function when a Consolidate dialog box opens in LibreOffice Calc? [CBSE Board Paper 1 (Set 4)]
(a) Average (b) Sum (c) Max (d) Var

Ans: (b) Sum

Q3. Consolidating data is useful for identifying: [CBSE Board Paper 2 (Set 4)]
(a) Errors and spelling mistakes (b) Trends and relationships (c) File size (d) Cell colours

Ans: (b) Trends and relationships

Q4. While consolidating data, a cell range can be named using _____ option. [CBSE Board 2024]
(a) Name range (b) Define range (c) Consolidate name (d) Define name

Ans: (b) Define range (accessible via Data → Define Range)

Q5. Rohit is analyzing sales data spread across four different sheets in LibreOffice Calc. He wants to combine the data into a single sheet to identify overall trends and relationships. He also wants any change in the source sheets to automatically update in the consolidated sheet. Which feature and option should he use? [CBSE SQP 2024-25 / 2025-26 / 2026-27]
(a) Use Pivot Table with 'AutoFilter' (b) Use Consolidate with 'Link to source data' enabled
(c) Use Data Validation with 'Cell Range' (d) Use Scenario Manager with 'Copy Results To'

Ans: (b) Use Consolidate with the 'Link to source data' option enabled

Q6. A teacher wants to prepare a summary report in LibreOffice Calc by calculating the average marks of students from two different spreadsheet files: X-A.ods and X-B.ods. Which feature of Calc will help her achieve this? [CBSE SQP 2025-26 / 2026-27]
(a) Data Consolidate (b) External References to other documents (c) What-if Scenarios (d) Goal Seek

Ans: (a) Data Consolidate (with function selected as Average)

Q7. What is the primary function of the Subtotal tool in Calc? [CBSE SQP 2024-25]
(a) It creates a chart for the data.
(b) It automatically groups data and applies common functions like sum and average.
(c) It highlights cells with conditional formatting.
(d) It removes duplicate data from the spreadsheet.

Ans: (b) It automatically groups data and applies common functions like sum and average.

Q8. _____ feature adds data arranged in a group of cells in Calc, with labels for columns and/or rows. [CBSE Board 2023]
(a) Average (b) Subtotal (c) Goal seek (d) Solver

Ans: (b) Subtotal

Q9. _____ feature of Calc is used to test 'what-if' questions. [CBSE Board 2023]
(a) Solver (b) Goal seek (c) Scenario (d) Styles

Ans: (c) Scenario

Q10. In LibreOffice Calc, the _____ tool is used to predict the output while changing the inputs and thus, one can choose the best plan of action based on it. [CBSE Board Paper 2 (Set 4)]
(a) Scenario (b) Subtotal (c) Consolidate (d) Relative

Ans: (a) Scenario

Q11. Radha is planning a school event and wants to decide how much money to allocate for different activities, like games, food, and decorations. She creates several different budget plans to see how changing the amount for each category affects the total expenses. How can Radha efficiently compare the different budget plans in the same spreadsheet? [CBSE SQP 2024-25]
(a) By writing down each budget on paper
(b) By using the What-if scenario feature in the spreadsheet to create and analyze different budget options
(c) By asking her friends to guess the costs
(d) By using the Filter function to look at one category at a time

Ans: (b) By using the What-if scenario feature in the spreadsheet to create and analyze different budget options

Q12. Ravi is analysing monthly sales data in Calc. He wants to test how changing the discount rate affects total revenue. Which feature should he use to simulate different outcomes? [CBSE Board Paper 1 (Set 4)]
(a) Sort (b) Indexing (c) What-if Scenarios (d) Filter

Ans: (c) What-if Scenarios

Q13. Ritika has created a spreadsheet to calculate the EMI for a car loan. The EMI depends on loan amount, interest rate, and duration. She fixes loan amount and interest rate, but wants the EMI to be exactly Rs. 12,000 per month. Which feature of Calc should she use, and why? [CBSE SQP 2024-25 / 2025-26 / 2026-27]
(a) Goal Seek - because it adjusts the input (loan duration) to reach the exact target output (EMI = Rs. 12,000).
(b) Scenario Manager - because it allows creating multiple possible outcomes with different sets of inputs.
(c) Consolidate - because it combines data from different sheets into one summary.
(d) Data Validation - because it restricts the type of values that can be entered into a cell.

Ans: (a) Goal Seek - because it adjusts the input (loan duration) to reach the exact target output (EMI = Rs. 12,000).

Q14. Goal Seek feature is available under _____ menu in LibreOffice Calc. [CBSE Board Paper 2 (Set 4)]
(a) Data (b) Tools (c) Insert (d) Functions

Ans: (b) Tools (Path: Tools → Goal Seek...)

Q15. _____ is specifically designed to minimize the results according to a set of rules that we define in a spreadsheet. [CBSE SQP 2023-24]
(a) Goal Seek (b) Scenario (c) Solver (d) Subtotal

Ans: (c) Solver

Q16. _____ is a more elaborate form of Goal Seek. [CBSE Board 2024]
(a) Database (b) Hyperlink (c) Report (d) Solver

Ans: (d) Solver

Q17. _____ can help to clarify the relation between cells while using 'Multiple Operations'. [CBSE SQP 2023-24]
(a) Detective (b) Solver (c) Navigation (d) Hyperlink

Ans: (a) Detective (Tools → Detective)

Part B: Short Answer Questions (2 Marks Each)

Q18. What is the purpose of consolidating data in spreadsheets? Discuss briefly. [CBSE Board 2024, Q.12]

Ans: The purpose of consolidating data is to combine, aggregate, and summarize data from multiple sheets or workbooks into a single master summary sheet while applying a function like Sum or Average. It enables users to view high-level summaries and analyze trends across scattered branch or departmental data efficiently.

Q19. Give any two prerequisites before consolidating data in Calc. [CBSE Board Paper 1 (Set 4), Q.13]

Ans:
1. **Standardized Row and Column Labels:** All source ranges must have identical headers and labels so Calc can match corresponding items accurately.
2. **Consistent Data Types:** Data within the corresponding rows/columns must follow the same format and numeric structure across all source sheets.

Q20. Briefly explain the purpose of the Subtotal tool in LibreOffice Calc software. [CBSE Board Paper 1 (Set 4), Q.12]

Ans: The Subtotal tool automatically groups data rows based on identical categorical values in a selected column, computes intermediate mathematical subtotals (like sum, average, count) for numeric data columns, and appends a final grand total at the bottom. It also creates an interactive collapsible outline view on the left margin.

Q21. How is Goal Seek different from Solver in a Spreadsheet? [CBSE SQP 2023-24, Q.14]

Ans: Goal Seek calculates backward to determine a single unknown input variable to achieve an exact known target output without constraints. **Solver** is an advanced optimization tool that deals with multiple unknown variables and complex operational constraints to find the combination that maximizes or minimizes the outcome.

Q22. Guneet has completed his assessments in four subjects and received the following marks: 75 in Math, 68 in Science, 82 in English, and 70 in History. The student needs to achieve an overall aggregate of 75% across all five subjects to meet his academic goals. He still has to take the exam in the fifth subject, which is Geography. Which Calc tool and menu option can he use to check how many marks he has to score in the 5th subject to get the required percentage? [CBSE SQP 2024-25, Q.14]

Ans:
• **Calc Tool:** Goal Seek
• **Menu Option: Tools → Goal Seek...**
(Setting: Formula cell = Average/Percentage cell, Target value = 75, Variable cell = Geography marks cell).

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Part C: Descriptive, Long Answer & Case-Study Questions (4 Marks Each)

Q23. Anshita is preparing spreadsheet notes for her Term Exam. Help her to write short notes on the following: [CBSE SQP 2022-23, Q.17]
(i) Scenarios (ii) Goal Seek (iii) Solver (iv) Subtotal

Ans:
(i) Scenarios: A what-if analysis tool in Calc used to test and store different sets of alternative input values (such as interest rates or sales forecasts) on the same sheet. Users can name them and switch between them easily.
(ii) Goal Seek: A back-solving tool that calculates backward to find the specific input value required in a single cell to achieve a desired formula target value.
(iii) Solver: An advanced form of Goal Seek used for optimization problems. It handles equations with multiple unknown variables and constraints to find the maximum or minimum possible outcome.
(iv) Subtotal: A feature that automatically groups rows of related data based on a chosen category and calculates intermediate totals/averages along with a grand total.

Q24. Explain the term 'Scenarios' in relation to Spreadsheets with an example. [CBSE SQP 2023-24, Q.18]

Ans:
A **Scenario** is a saved set of cell values for calculations within a what-if analysis model. It allows users to simulate multiple possibilities without creating separate spreadsheet files. Each scenario is given a distinct name and can be quickly chosen from a drop-down list on the sheet.
Example: While calculating a car loan EMI, a buyer can create two scenarios: *'Loan_4Years'* (48 instalments) and *'Loan_5Years'* (60 instalments). By switching between these scenarios from the scenario header drop-down, the buyer can instantly compare how the monthly EMI and overall budget differ between the two payback periods.

Q25. Case Study: Mr. Rakesh owns a small grocery shop. One of his friends suggested using LibreOffice Calc, which includes What-if Analysis tools such as Multiple Operations. This will help him manage the details of his shop's stock and analyze the stock data effectively. Help him understand by answering the following questions: [CBSE Board Paper 2 (Set 4), Q.18]
(a) What is the use of What-if Analysis tools?
(b) What is the use of the Multiple Operations tool in LibreOffice Calc?
(c) Select the _____ menu → Multiple Operations to use the Multiple Operations tool.
(d) Multiple Operations tool uses _____ (one / two / three) arrays of cells.

Ans:
(a) What-If Analysis tools are used to test different input values in cells to see and forecast how those changes affect the outcomes of formulas in the worksheet.
(b) The Multiple Operations tool generates a formula array table that calculates and displays results for multiple alternative input values simultaneously in a side-by-side comparative grid.
(c) Select the **Data** menu → Multiple Operations (Path: **Data** → **Multiple Operations...**).
(d) Multiple Operations tool uses **two** arrays of cells (one array of values in a column and another array of values in a row when evaluating dual-variable formulas).

A.S. NOTES IT 402